

Design and Implementation of a Semi-Autonomous Unmanned Ground Vehicle

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Abstract

This paper presents the design, control and implementation of a generic wireless-operated semi autonomous unmanned ground vehicle (SUGV). The compact and lightweight robotic platform provides a safe and effective means to deal with hazardous-duty operations. The configuration of the vehicle system includes robust and easy to transport mobile base unit housing a single board computer for the vehicle control, and a command and control unit which consists of windows based commercial laptop loaded with the robot's software. A wireless network card provides an operator the ability to manipulate and control the SUGV remotely from a safe location. The operational capabilities of the vehicle system include remote reconnaissance, surveillance, communications relay and the ability to interface peripherals supporting different mission requirements. Design considerations, control algorithms, and the experimental results achieved through the prototype vehicle are focused. The significance of this work is in the increased understanding of remote robotic response and the applications of unmanned ground vehicles for industrial and military operations.

I. Introduction

There are a variety of potential applications of robotics to land operations that can increase efficiency and safety. These include reconnaissance, target engagement, airport security, minefield detection and neutralization, explosive ordnance disposal, physical security and operations in contaminated environments [1, 2, 3]. The rising industrial demand has led to rapid development in this field and has become a major challenge in robotics research and development sector. This paper mainly deals with the design and implementation of a prototype vehicle for unmanned operations.

The generic structure of this vehicle consists of two parts coupled by a communication link based on a 2.4GHz

wireless LAN [4, 5]. The mobile base unit (MBU) consists of a flexible platform to handle diverse and numerous actuators. The vehicle control architecture consists of an embedded single board computer allowing the use of latest operating system versions as well as the ability to interface peripherals based on current industrial standards. A charge-coupled device (CCD) camera [6] mounted on pan and tilt platform with remote focus and zoom and a front ground level camera provides a total field of view. A two way digital audio is added for hostage negotiations and communication at a distance. The off vehicle command and control unit (CCU) will issue commands to the on-vehicle controller. The server system will be controlled by a human/operator through a laptop keypad or a joystick [7].

II. Mechanical system

A. Design criteria

The basic task was the development of a generic platform with emphasis on compactness, durability and simplicity. The platform had to be capable of negotiating outdoor environment: multi-terrains, small obstacles, little acclivity and movement in shallow water. The platform also needed to take a load of about 20 kg in form of additional devices (for specific tasks). The platform had to weather different climates, so whole of control unit including batteries and motors required proper housing. The fabrication of such rugged and at the same time lightweight and easily transportable vehicle was an uphill task.

B. Vehicle structure

Standard design procedures were used for initial calculations [8, 9]. The design software used are

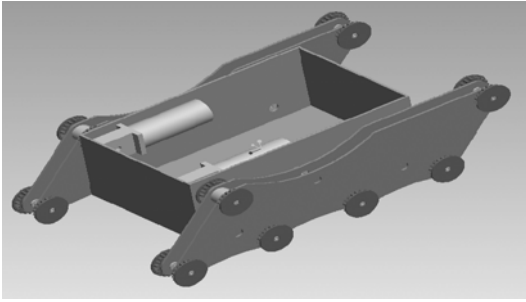


Fig. 0 Pro-E model of the platform.

AutoCAD 2000 for drawing and Pro-Engineer Wildfire for simulation. The structure of the SUGV can be divided into two sections; the chassis and the driving mechanism. The chassis serves as the housing for motors, batteries and the on vehicle controllers. Attached to the chassis is the driving mechanism, which includes motors, driving wheel, the idlers and the caterpillar tracks (in Fig. 2) and the additional mounting platform to shoulder mission specific equipment (that may be a robotic arm or bomb disrupter).

The material used for bulk of the structure is Aluminum Alloy **AL-6061-T6** [10]; mild steel has been used specially for axels and shafts where the greater strength is required in a small volume.

Two independent DC servomotors make up the single axis differential drive of the vehicle. The idlers provide more contact area with the ground, which in turn makes the movement more durable. The tracks are made up of synchronous rubber timing belts: **850H** having a circumference of 85 inches and standard 0.5H teeth. The idlers and driving wheel are made of aluminum with their teeth specifically designed for 0.5H teeth. A flexible feature added to the drive mechanism is that it can be easily removed from the chassis by loosening three fasteners.

C. Motors selection

The major issue for mobility of platform is of motors

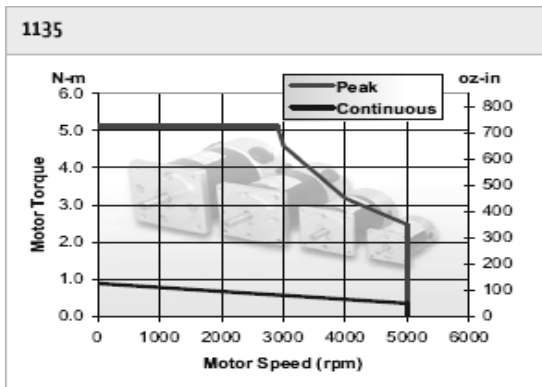


Fig. 2 Torque-Speed curve of driving motors [12].

which has a few selection parameters. Firstly the **weight** of the SUGV needed to be minimized in order to consume less power. Approximate weight of structure: the chassis and drive comes out approximately 35 kg, which has been symmetrically distributed. Second crucial parameter is the vehicle **speed** and it has been limited to 0.30 m/s. Third parameter is the **acceleration time** which has been selected as 1 sec. Based on these assumptions the power requirements were calculated [11] and the operational performance parameters are tabulated in Table 1.

$$P_{out} = \tau \times \omega$$

$$\text{Power for climbing} = \text{Mass} \times g \times V_v$$

$$\text{Power needed to accelerate} = KE / \text{acc. Time}$$

$$\text{Here } V_v \text{ is the vertical velocity} = \text{speed} \times \text{Gradient}$$

TABLE I

Performance Parameters

Parameters	Operational Performance
Max. Vehicle Speed	0.30 m/s
Rotational Speed (ω)	65 rpm
Total mass	55 kg
Gradient climb	30 %
Total power reqd.	120 W
Operation Voltage	24V

To meet the above performance requirements, DC motors coupled with a gearbox of 1:20 to give a speed of around 1300 rpm at 24 volts.

The torque speed curve of the motor is shown in Fig. 3. The worm gears precluded the platform from the use of braking mechanism [5].

III. System Architecture

The Mobile Base Unit (MBU) is integrated to the Command and Control Unit (CCU) through a 2.4 GHz wireless LAN installed on a commercial laptop. The complete software for vehicle control has been implanted on Windows based environment. The control board is based on a Single Board Computer (SBC) that is connected to the MBU subsystems through a RS232 serial port. Fig. 4 shows the onboard and off board electronic components integrated to provide complete command and control for the SUGV platform.

IV. MBU Controllers

A. Single Board Computer

The compact command and control counsel come from a Single Board Computer with an embedded ultra low

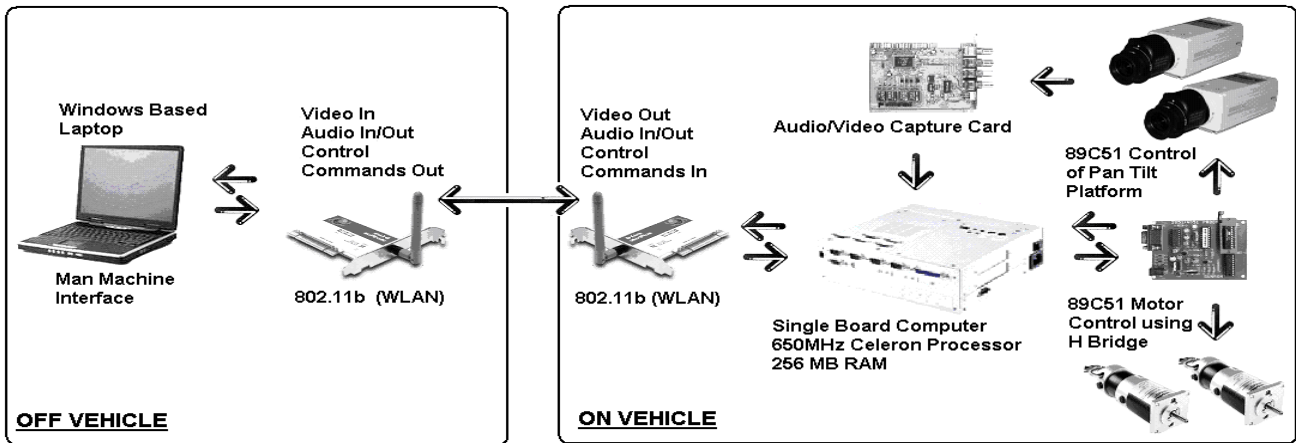


Fig.3 System Architecture.

voltage Intel Celeron 650MHz Processor designed on the EBX form factor that supports the PC/104+ interface for ISA/PCI module upgrades. The board not only offers a variety of space hardened characteristics but allows the use of latest operating system versions as well as the ability to interface peripherals based on current industrial standards [13]. A PC/104 vehicle power supply with a 50-Watts output and an input voltage range of 6 to 40VDC supplies power to the SBC.

On-board features include a PCI slot, an LPT, 2 USB ports, IrDA, 4 serial ports (three RS-232 and one RS-232/422/485) and audio interface.

The SBC served as an ideal central computer for the operation of MBU under harsh environmental conditions and added to the generic nature of the platform as the wide range of ports can be used for controlling other mission specific peripherals.

B. Motion control system

Motion Control is a critical factor in the design of a robot. Separate control systems integrated under a central computer enables to drive MBU with the simultaneous control of additional on board devices. The motion control mechanism is focused to drive the platform to the desired direction at a desired speed and perform the task called for. An operator gathers the information from the on-vehicle cameras and gives commands that are wirelessly received by the SBC, which is connected to motion controller based on Atmel 89C51 microcontroller (μ C).

a. Motion controller hardware

The motor drivers used are a full bridge MOSFET design giving the capability of a four-quadrant control [14]. The gate driver circuitry consists of a HIP4081A monolithic full H-bridge driver chip that includes both high- and low-side FET drivers and all needed voltage boost circuitry. The 89C51 is used as the digital signal source to provide the PWM signals into the 4081A inputs

in order to drive the bridge [15, 16]. The pan, tilt, zoom and focus control for the on board cameras is also provided though the μ C board. These are then interfaced through the RS232 port of the controller board to the SBC.

b. Motion Behavior Programming

The motion control system is programmed to provide forward, backward, left and right movement to the platform with the ability to increase or decrease speed as well. To provide maneuverability with sharp turns motion controller is programmed to make turns at zero radiuses. The μ C control program is written in C language using Keil μ Vision software.

V. Command and Control Unit

A commercial laptop serves as the CCU running on Windows environment. The laptop is loaded with the robot's software, which shows the video input of two on board CCD cameras and provides a multifunctional control for the MBU and its additional peripherals. Also if a microphone and speaker is attached, a two way audio can be enabled. An operator can control the platform through a laptop keypad or a joystick.

VI. Software development

The software is designed under Windows using Microsoft Visual C++. The software consists of three portions; the man machine interface, the embedded system software and the communications channel.

A. Man machine interface

The robot software consists of a very friendly graphical user interface [17], making the control of the vehicle easy for even a layman. The operator can issue any of the following control commands for the vehicle:



Fig. 5 Graphical User Interface.

- Maneuver the vehicle in any direction
- Increase or decrease speed of the vehicle
- Control the pan /tilt of the drive camera and adjust its focus and zoom
- Capture still frames from any of the two video streams
- Enable the sound channel for two-way audio communication
- Shutdown or restart the onboard controller

The software can be controlled via mouse, keyboard or a joystick.

B. Embedded system software

Embedded system software development practices are in many ways different from other software engineering disciplines [18]. The software for the SBC is designed under Windows as it met the real time constraints for the application just as well. Microsoft Visual C++ was chosen due to its flexibility. Video capture and sound capture are done using the Video for Windows and Windows Multimedia [19] libraries respectively.

The video capture and transmission has been executed as a separate thread, which is created as soon as the SBC boots. On the event of reception of a command from the CCU, the controller immediately interprets the command and issues a corresponding command to the motor controller through serial port.

C. Communication with Hardware Ports

For the purpose of data transfer to move the vehicle, the SBC has been connected to 89C51 μ C via serial port using RS232 standard. The serial port has been accessed using the file input and output (I/O) functions (CreateFile, CloseHandle, ReadFile, ReadFileEx, WriteFile, and WriteFileEx)[21]. A baud rate of 9600 has been set for both transmission and reception.

D. Communications Channel

The wireless communications link is established between the mobile base unit and the command and control unit on enhanced IEEE 802.11b standard, configured for peer to peer communication. A derived class of the CASyncSocket [22] is used to form the sockets for communication in the program. The channel transfers audio, video and data for the operation of SUGV. Video and data are transferred on the same port, whereas a dedicated port is used for the sound transfer.

If the vehicle is within a range of 75 meters of the CCU, 97.01% of the information sent is received correctly at the CCU. However if the distance is increased farther than that the communication channel becomes weaker.

VII. Results

The prototype platform has proved to be robust during test operations under various environments such as acclivity, shallow water, and sedimentary surface with different loads. The weight distribution and use of caterpillar tracks enabled the platform to climbing stairs and slopes inclined at about 30 degrees. The generic structure of the MBU controllers provided easy integration of additional devices based on currently accepted industrial standards. The designed software for the CCU provided easy control and multifunctional capabilities to control vehicle and other peripherals. The use of wireless LAN for the prototype vehicle provided a large data transfer rate.

VIII. Summary and Conclusions

An integrated version of rugged mobile platform is presented by building up a lightweight tracked vehicle. Various features of the design and control of unmanned operation are experimented. The MBU has provided a generic test bed for future study and enhancement of UGV system. The MBU controllers provided a platform to integrate and test the remote response of additional devices such as a robotic arm or a bomb disrupter. The current research includes the implementation of various image processing algorithms to avoid obstacles.

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